



MIDDLE EAST TECHNICAL UNIVERSITY  
NORTHERN CYPRUS CAMPUS

DEPARTMENT  
OF  
COMPUTER ENGINEERING

---

**CNG 351**  
**Data Management and File Structures**  
**Assignment 1**  
(6% of actual grade)

---

**DUE DATE:** 1, November 2024, 23:55 (Cyprus Time)

## IMPORTANT RULES

1. Use a graphical tool such as Lucidchart or Draw.io.
  - Lucidchart can be used to design and create your solution. You can find a full tutorial on ER diagrams and Lucidchart here:  
<https://www.lucidchart.com/pages/er-diagrams>.  
Please make sure that you use **Chen notation** for your design, other notations will not be accepted.  
For Lucidchart, please ensure that you have an educational account. **To get an educational account, you need to sign up with your METU mail addresses from the following link:**  
<https://lucid.app/pricing/lucidchart>
  - Draw.io can also be used to draw your diagram. You can find the full details here: [www.draw.io/](http://www.draw.io/).
2. Any assumptions must be explicitly explained. If your assumptions do not make sense in the given requirements content or conflict with the requirements or similarly, if you assume and did not specify the assumption, you will lose points from that part of your design. Please do not include trivial assumptions; for example, “we assume id is unique”.
3. Create a PDF file for your solution and upload only one PDF file to ODTUClass, one team member is enough to upload the solution.
4. Assignments will be completed by a team of two/three people. Your team details are recorded on odtuclass, please note that you **cannot** change your team members.
5. Please submit a report that includes a cover page with the team details, including your team/group Id, group/team member names/surnames and their student IDs.

## PURPOSE

The main purpose of this assignment is to review the modelling of a relational database from requirements using Entity-Relationship Diagrams (ERD).

## GRADING

This assignment includes four use cases for different database applications. The data requirements of each of these use cases need to be modelled with ERD, and the grading will be as follows:

1. Use Case 1: 22 points;

2. Use Case 2: 23 points;
3. Use Case 3: 25 points;
4. Use Case 4: 25 points;
5. Please note that you are not designing the application; you are designing the **data requirements** of the application.
6. Your designs are expected to be clean and readable. You will also be penalised if you use ternary (or n-ary) relations. Please make sure that all your designs consider cardinality and participation constraints.
7. Report: 5 points. A good report means typewritten, complete (every section fulfilled), with clear explanations in English (where relevant), and submitted via ODTUCLASS as one combined PDF document. The report's first page must be a title page that clearly states team details and assignment number. Each use case should also include *Assumptions* written. The footer of all subsequent pages should be numbered in the format x of y (e.g. 2 of 6), etc.

## USE CASES

### Use Case 1 - METU NetRegister

At METU NCC, students must register their devices to our campus network registration system to gain access to the campus network. The university has implemented a comprehensive network registration system called METU NetRegister to manage device registrations and facilitate this process. The main goal of this application is to store information about students, devices, registrations, network engineers, and related network issues and keep track of the registered devices' VPN connection details. The details of the system are as follows:

- First of all, the system will keep track of the students and their details. The system will keep track of a unique student ID, name, surname, email, phone number, password, and address for each student.
- To be able to access the Internet, students need to register their devices, which can include laptops or phones. Each device has its unique IP address, model, type (laptop or phone), and specifications (e.g., operating system (OS), OS version, memory).
- The registration of each device will be tracked, and the system will store details such as the registration ID, time and date of the registration, which will be recorded as a timestamp, and registration status (e.g., pending, approved). The locations where the student intends to use the network (e.g., library, dormitory, or classroom) will also be recorded. This allows the university to understand where students are accessing the network. The connection type (e.g.,

wired or wireless) will also be recorded during registration, ensuring that the university can effectively manage its network resources. Many devices a student owns can be registered, allowing the system to know which student owns which devices. However, once a device is registered, the same device cannot be registered again.

- This registration system will also maintain IT personal details that will actively monitor the registrations and issues that can arise. These IT personnel are called Network Engineers. The university currently has four network engineers. The network engineers can approve or decline a registration. This is recorded in the system. Therefore, it should be known which network engineer approved a device registration. The network engineers can also block a registration on the system. When registration is blocked, the system needs to track which network engineer blocked it, when (day/time), and for how long. The system keeps track of such blockings regarding the number of days. They can be blocked for 3/5 or 7 days. The system maintains a unique ID, name, surname and email for each network engineer.
- The system will record any network issues that also arise. One specific registration can have multiple issues. The system also keeps track of these issues. This allows for efficient troubleshooting and tracking of recurring problems, as each issue will have a description, timestamp when the issue occurred, and status (e.g., resolved, unresolved) recorded. Please note that two issues related to one registered device cannot occur at the same timestamp. But of course, many issues can occur for a registered device. Each issue is also assigned to a network engineer in the system. This ensures that every issue is properly and timely addressed.
- Finally, once devices are registered, they can also have VPN connections. These VPN connections are costly; therefore, the system also keeps track of the VPN connections. For each connection, the system records when the connection started and stopped. Please note that at a given time, there can be only one connection for a registered device, but of course, a registered device can have many VPN connections.

## Use Case 2 - ACM Fest Management System

METU NCC ACM Student Society organizes a hackathon annually in the name of “ACM Fest”. The society developed an online ACM Fest registration system to keep everything organized and create an archive for every year’s submissions and details. Anyone participating in these annual hackathons, including supervisors and judges, should use the system.

- The participants should register in the system before assembling a team. They must enter their email, name and surname, department, date of birth, and university. Department and university information is recorded as most participants

are expected to be university students. After creating a participant profile, they can create a team to participate in hackathons. A student can create multiple teams but can not be a member of more than one team simultaneously. (They can make different teams for every different hackathon.)

- In the hackathons, the team can include between one to five participants. Every team has an ID, a name, a catchword, and if it represents a university, that is also recorded. Also, teams can add their supervisors if their supervisors have been registered before. Supervisors should register to the system with their email, name and surname, profession (e.g., faculty member, software engineer, etc.), and a list of their research areas. Supervisors can supervise multiple teams at the same time.
- After the registrations and team creations are done, teams will add their projects for that year's hackathons into the system. A project can not exist without a team. Team members will add their project name and description to the system. Note that project names can be the same within different years from different teams, but the same team can not use the same project name. Every team can participate in only one project per year, but if they don't disband the team, they can create another project for the next year's event.
- During the hackathon, the projects will be evaluated by that year's judges. The society organization team will send the special registration link to the academics and industry veterans if they want to become judges for the first time. They must enter their name and surname, email, gender, profession, and date of birth. After registration, the system will create an ID for judges, who will be added to the system for the current and next hackathons.
- Judges will evaluate the submitted projects of that year's hackathons with a score out of 100. A project is evaluated by at least four judges. However, every judge is responsible for evaluating every single project that participated. The final scores of the projects will be the average of every score the judges gave.
- The system also will keep track of the hackathons since every year's event is different. Every hackathon will have its ID, year, theme, and duration. Please note that the system needs to record which projects are submitted for which hackathon. Plus, every year, there can be only one winning project in that year's hackathon. The winning project also needs to be recorded.

## Use Case 3 - Desire for Velocity

Desire For Velocity is a multiplayer competitive racing game developed by some brilliant indie game developers in METU NCC. As a candidate computer engineer, you have been selected to work on the project as a database designer. You need to design a database that will fulfil the requirements of an online racing game. It's a real-time

racing game with different cars and car classes, races with matchmaking, tournaments, and single-player drift mode.

Every player will have their ID, nickname, and password for registration and authentication. The system will also keep track of their MMR score for proper matchmaking in the tournaments and the credits for their wallet so the players can buy new cars. There will be lots of cars with different classes in the game. Every car will have its brand, model, and color. When a player buys a car, he or she must name the car to own it. Different players can have same-name cars, but a player can not name more than one car with the same name.

Players will join multiplayer races with their cars. Every race will have an ID, map, race type, and the number of laps. Single races will be non-competitive, and players are free to choose any car they own in every different race. The database should keep track of the player's position and the duration of every race.

For the players who want competition and want to be matched with their own MMR level, the game will have a tournament system. Tournaments have multiple races in it. Every tournament will have its ID and an MMR range for that tournament. Since it is competitive, the players will choose a car at the beginning, and they have to finish all the races in the tournament with that car. In other words, tournaments are a set of races that should be recorded as a group in a database. When players enter and play tournament mode, they will be playing a set of races part of the tournament.

The game will also have a single-player mode, which is Drift Runway. It will be a different game mode than Races, where the players will play it alone with a dedicated car for that drift game. That means players cannot select their car but the system uses its own car. All players should use the car that is chosen by the system and they cannot change it. This is to ensure that the game is fair to all players. This system car information is not recorded in the database as it is custom generated for each drift runaway. Every drift round has its ID and map. Also, the player's score for every round they play should be recorded as well. If a player scores well and makes the top 3, the player can leave a quote on that drift runaway. Three quotes from three players who score top three will be displayed at the beginning of the game.

During the races, if the same players encounter each other more than once, and one of the players outperforms the other more than a certain time, they will be assigned as nemesis of each other. Every player can become a nemesis of multiple players, and this information will also be recorded in the system.

## **Use Case 4 - Online Tutoring System**

In Cyprus, a smart online tutoring system was developed to address the growing demand for students for flexible and additional academic support offered by tutors

who specialize in various subjects. This platform connects students with the tutors in a way that allows them to conduct private tutoring sessions either for supplementary help in their ongoing courses or to gain proficiency in a new subject. The system integrates functionalities other than connecting students with tutors, such as progress tracking, course and lecture management, and secure payment processing.

For students to be able to access the system, they need to register by providing their personal information, such as their student ID, name, surname, email, phone number, and university that they are currently studying.

Like students, tutors register on the platform and provide detailed profiles, including their name, surname, email, and areas of specialization, and the system will automatically generate a unique ID for each tutor. These profiles allow students to search and select tutors who meet their specific learning needs. Tutors should also have a rating that is provided by students after each lecture. This ensures the quality of the tutoring experience. Each tutor teaches several courses that have a unique ID, a description, and a name, recorded in the system. For example, a possible course would be C-Course or Java-Course. Moreover, once a course is added by the tutor, they should add the lectures associated with that course. For example, a lecture would be from Monday 10-11.

Once a student and tutors register to the system, students can view the profiles of the registered tutors, their courses, and their related lectures. They will also be allowed to get feedback on their progress from the tutors after each lecture, and provide feedback or rank their tutors based on their learning experience after the lecture.

Once students book a lecture, they are assigned a lecture session that a tutor has already scheduled for a specific course. Each lecture has an identified cost, date, start time, and end time. Moreover, the system ensures secure payment handling and records all completed transactions. Thus, once a lecture is booked, a payment process is initiated, where students must pay using various methods. The payment has a unique ID and records of the date, time, amount, and payment method.

As mentioned earlier, when the lecture finishes, tutors are responsible for generating and sending progress reports to the students. These reports provide detailed feedback on the student's performance and progress. Students, in return, receive these reports, which allows both the student and the tutor to track the improvements over time. In these reports, tutors record the feedback that they want to give to the students, for example, if the student was able to follow the lecture and if they were able to understand the lecture, the system also records the date and time of the report. Similarly, after each lecture, students can rate their tutors based on their performance. For ratings, the system uses a rating between 1-10 (1: bad, 10: excellent). These ratings are reflected in the tutors' profiles so that they help other students select tutors; it also encourages the tutors to deliver high-quality education and improve their skills based on students' evaluations. In the tutor profiles, the

system calculates the averages of all the ratings they receive for all their lectures. Therefore, if a tutor teaches two courses with two lectures each, their overall rating will be the average of all four lectures.

Furthermore, the system has a number of administrators who manage the registration of the tutors; their job is to check the tutor's credentials. This ensures that the tutors on the system are well-qualified and provide quality tutoring. A tutor applies and once they are approved, then the system records who approved a tutor. For admins, the system records, their admin ID, username, name/surname, and email.